

OBA311 Business Analytics I

Fall 2024 | CRN24001 MW 10am | Chiles 225

Instructor	Shailesh J. Divey	Phone	617-480-7857 (Mobile, txt msg OK)
Office Hours	Tue/Thurs, 10-11 am via Zoom at Zoom Office , or at Lillis 474 in-person meeting by appointment.	E-mail	sdivey@uoregon.edu Include OBA 311 and section in the subject

- All the communication throughout the term will be through **announcements in Canvas**. Therefore, you must **regularly check the announcements** to be informed about the course.
- All times mentioned in the syllabus and throughout the term are in **Pacific Standard Time (PST)**.

Overview of the Course

This course provides undergraduate business students with the fundamental concepts and tools needed to understand the emerging role of business analytics in organizations, apply basic business analytics tools in a spreadsheet environment, and effectively use and interpret analytic models and results for making better business decisions. Over the quarter, we focus on the following topics:

- The basic foundations needed to understand business analytics and manipulate data;
- The fundamental tools and methods of data analysis and statistics including but not limited to data visualization, descriptive statistical measures, probability distributions and data modeling, sampling and estimation, building and analyzing models on spreadsheets, and simulation and risk analysis.

Required Materials and Tech

Textbook

Business Analytics, 3rd edition, by James R. Evans. Pearson, 2019 (ISBN: 9780135231678).

The textbook is available in DuckStore. The 2nd edition is fine too if you can find it for less expensive. Note that the homework assignments are based on the 3rd edition. It can also be purchased or rented via online retailers. You will use the same textbook for Business Analytics II. Textbook datasets are available to students at [Pearson Online Resources for Business Analytics](#).

Software

- This course extensively uses **MS Excel** and **MS Access** (for database management and SQL).

How Your Grade Will Be Determined

Assessment Relative Weight (Total 100%)

Participation 10%

Homework assignments 15%

Quizzes 15%

Midterm Exam 30%

Final Exam 30%

TENTATIVE SCHEDULE* Below is a tentative schedule of this class.

Topics

- 1 Intro. to Business Analytics
- 2 Data Visualization
 - [NYT Learning Network - What's Going On in This Graph?](#)
 - **Discussion** - Groeger, L. "[Design Principles for News Apps & Graphics](#)." *ProPublica*, May 30, 2013.
- 3 Descriptive Statistics
- 4 Probability Distributions and Data Modeling
- 5 Sampling and Estimation
- 6 Simulation and Risk Analysis
 - **Monte Carlo Simulation, Forecasting and Decision Trees**
- 7 Decision Analysis
 - **Uncertainty, Risk, Sensitivity, Value of Information, Bayes Theorem, Subjective Probability, Utility, Multiattribute Choice, Biases in Judgement**
- 8 Database Management and SQL
 - **Database Management Lifecycle - business rules, ER diagram, and normalization**
 - **Relational Database, SQL**
 - **Introduction of XML database (NoSQL database)**
- 9 Data Visualization - Unlock the Potential of Generative AI
 - **Understanding Training Data:** Identify biases and anomalies in data, and ensure that the data accurately represents real-world problems for generative AI.
 - **Performance Evaluation:** Visualize generative AI outputs to detect errors or artifacts, and enhance the training process and make adjustments to the model architecture as needed.
 - **Communication of Results:** Craft informative and visually compelling visualizations, and assist others in comprehending the insights derived from generative AI models.
 - **Debugging:** Visualize outputs at various training stages, and identify and rectify errors in both the training data and model architecture.
 - **Interpretation of Predictions:** Utilize visualizations to discern key features influencing predictions.
 - **Model Development:** Employ data visualization to investigate relationships between different features, and streamline the development of new generative AI models.